

Building a Geosphere

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Planar graphs

Drawing the planar graph of a polyhedron is a useful way to study its structure. In particular, it makes it easy to count the number of faces, vertices, and edges.

To obtain a planar graph of a polyhedron, imagine cutting open one face of the polyhedron and unfolding it onto a table so that no edges cross.

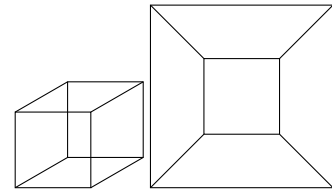


Figure 1: A cube and its planar graph.

Task 1

Draw the planar graph of an icosahedron. Make sure it has 20 triangles.

From the icosahedron to the football

We begin with an icosahedron. By subdividing each triangular face into smaller triangles and then projecting the vertices onto a sphere, one obtains a geodesic polyhedron.

One famous example of a polyhedron made from pentagons and hexagons is the football (truncated icosahedron).

Before studying the football, begin with the dodecahedron.

Task 2

How many vertices and edges does the dodecahedron have? How many vertices and edges does the football have? Find a relationship between the number of vertices V and the number of edges E that holds for the football. Does this formula also hold for the dodecahedron? Conjecture a formula that might be true for any polyhedron made entirely from pentagons and hexagons.

Let F_5 be the number of pentagonal faces, F_6 be the number of hexagonal faces, and F denote the total number of faces.

Task 3

Find F in terms of F_5 and F_6

Task 4

Each edge belongs to exactly two faces. Use this to derive a formula relating F_5 , F_6 , and E .

Euler's formula

Recall Euler's identity for convex polyhedra:

$$V - E + F = 2 \quad (1)$$

Use your answer from the previous tasks together with Euler's formula to derive a relationship between F_5 and F_6 .

What does your result tell you about the number of pentagons in any polyhedron made entirely from pentagons and hexagons?

Show that every polyhedron whose faces are only pentagons and hexagons must contain exactly 12 pentagons.

Building a GeoSphere

In this activity, we will construct a *frequency-6* geodesic dome. Start with an icosahedron and divide each edge into 6 equal parts. Joining the new points creates a triangular grid on every face.

Task 5

After subdividing each edge into 6 equal parts, how many small triangles does each original triangular face contain?

Around each original vertex of the icosahedron, five small triangles meet, giving rise to a pentagon. Around every new interior vertex, six small triangles meet, giving rise to a hexagon.

Task 6

How many pentagons and how many hexagons does this frequency-6 construction produce?

We will build the geosphere using bamboo sticks. Each edge of the final structure is formed by two bamboo sticks. Each bamboo stick is marked with a letter label. Figure 4 shows the labelling scheme. Use these labels as a guide while weaving the bamboo sticks together to assemble the geosphere.

Compare the geosphere you have built with the football studied earlier. What similarities and differences do you notice?

References

Geosphere at icts, December 2023. URL <https://www.icts.res.in/outreach/other-events/geosphere2023>.

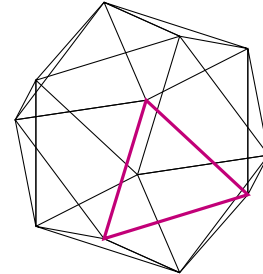


Figure 2: icosahedron

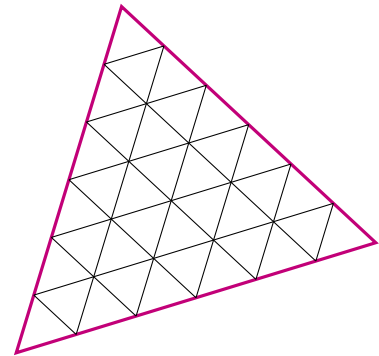


Figure 3: One triangular face of the icosahedron subdivided into smaller triangles.

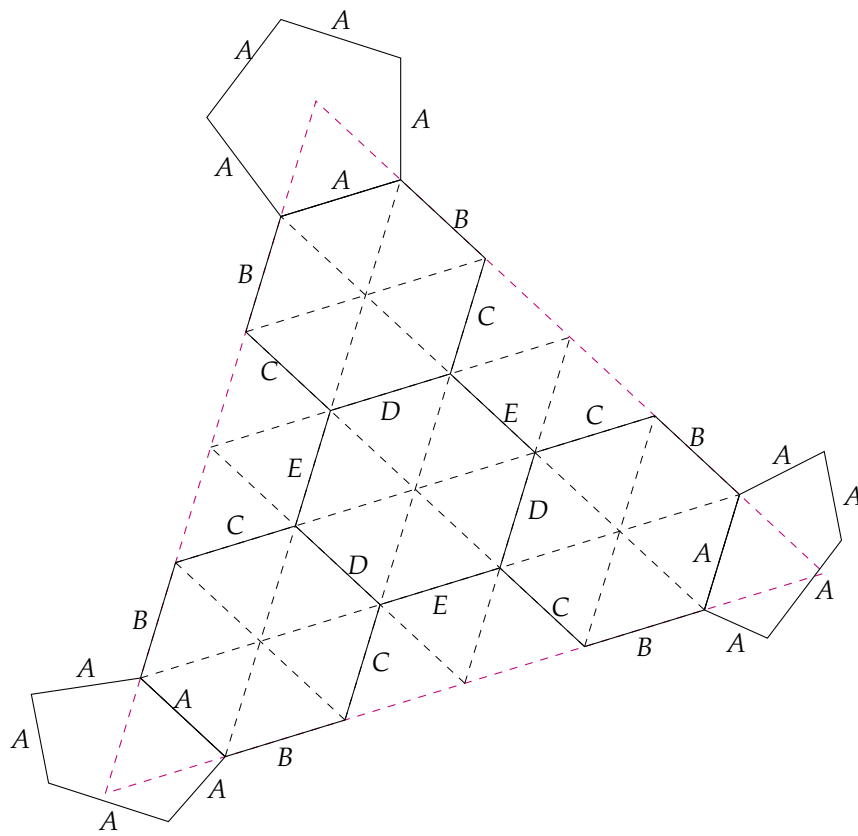


Figure 4: Schema